

[I225] Statistical Signal Processing(E) Office Hour 1

1. Imagine you are looking at the digital thermometer outside the university. We want to model the temperature recorded there.

Let $T(t, \omega)$ be the stochastic process of temperature at time t (hours past midnight).

The randomness ω comes from which specific **day** of the year we are looking at (e.g., April 1st vs. October 1st).

- A. Classify the following events: (constant value, random variable, stochastic process, or realization)

[1] The predicted temperature curve for tomorrow, from 0:00 to 23:59.

[2] The temperature measured at exactly 8:00 AM across every possible day of the year.

[3] The collection of every temperature reading ever taken at this station at every second of every day.

[4] The temperature at exactly 12:00 PM on January 1st, 2024.

[5] A value representing the average yearly temperature of the city.

- B. Based on how temperature behaves throughout a 24-hour cycle, would you classify the temperature process $T(t, \omega)$ as Wide-Sense Stationary (WSS), Strict-Sense Stationary (SSS), or Non-Stationary? Justify your answer using the **average** of the process.

- C. Suppose you want to find the average temperature of the city. You have two choices:

a. **Time Average:** Record the temperature of **one specific day** and average it over 24 hours.

b. **Ensemble Average:** Measure the temperature at **exactly 12:00 PM** across 365 different days and average those values.

Will these two averages give you the same result? Based on this, is the process **Mean-Ergodic**?

2. Consider a signal that has a constant DC offset A plus a random oscillating component:

$$X(t) = A + \cos(\omega_0 t + \theta)$$

- A is a constant.
- θ is a random variable uniformly distributed between $[0, 2\pi]$.

A. Calculate the average: $E[X(t)]$

B. Calculate the Autocorrelation: $R_{XX}(t_1, t_2)$

C. Calculate Autocovariance: $C_{XX}(t_1, t_2)$

3. You are designing a buffer for a video streaming app. The video requires **100 MB** to start playing. Because of a bad Wi-Fi connection, the download progress (measured in Megabytes or MB) behaves randomly. At every second:

- **Success (50% chance):** You download **1 MB**.
- **Buffering (50% chance):** You download **0 MB** (stay in place).

Let $X[n]$ be the total MB received after n seconds.

- Your app shows a progress bar to the user. Based on the **Ensemble Mean** $E[X[n]]$, after how many seconds should the app tell the user the download will 'likely' be finished?
- Using only the ensemble mean to predict the finish time results in a 'premature notification' 50% of the time (the app says it's done, but the file is still buffering). To provide a more reliable estimate, we want to find the time n where even an 'unlucky' user—defined as being one standard deviation ($\sigma_X[n]$) below the mean progress—has finished the download.

(Task: Solve for the time n required to reach the 100 MB target such that $E[X[n]] - \sigma_X[n] = 100$.)

4. You have a noise signal $X(t)$ that represents the background hum of a microphone. The probability density function (PDF) of this noise is **Uniform**, meaning it is equally likely to be any value between **0 and 1** ($f_X(x) = 1$ for $0 \leq x \leq 1$).

You pass this signal through an amplifier (a memoryless system) that doubles the signal:

$$Y(t) = 2X(t)$$

- What is the new range of the output $Y(t)$?
- As the total probability must still equal 1, but the range has changed (from question A), what must happen to the “height” of PDF ($f_Y(y)$) ?
- Use the transformation formula to find $f_Y(y)$:

$$f_Y(y) = \frac{f_X(x)}{|J(x)|}$$

where Jacobian $J(x) = \frac{dy}{dx}$